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Driving HSE Excellence: Instantaneous SimOps Visibility Through Next-Generation Technology for Safer, Smarter, and Simpler Multi-Site Execution

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Abstract

This paper introduces a scalable, Artificial Intelligence (AI) powered interface developed using Microsoft Power BI to manage Simultaneous Operations (SimOps) in high-density activity oil and gas environments. The tool enables real-time visualization of over 20,000 unique activity combinations across geographically distributed facilities. It provides actionable insights through a branched decision tree and centralized dashboard architecture that allows users to view, assess, and interact with key actionable operational data.

Developed as part of the Hail and Ghasha (H&G) mega project, this solution is a leap forward in digital SimOps coordination with reference to Facilities Engineering, Procurement and Construction (EPC) works, Drilling works (rig and well testing) in limited space such as the offshore drilling centers. It enhances operational decision-making by identifying permitted and non-permitted operations and aligns with Upstream 2.0 – ADNOC's digital transformation vision for upstream oil and gas operations. The tool is built on a smart, structured framework incorporating activities, sub-activities, restriction types, and Health, Safety, and Environment (HSE) requirements. These are dynamically processed by AI-enhanced Data Analysis Expressions (DAX) logic within Power BI, facilitating intuitive category-based filtering, restriction awareness, and proactive safety measures.

The interface functions smoothly on intrinsically safe smart devices, equipment designed for use in hazardous areas, preventing ignition of flammable gases or vapors, delivering SimOps insights during walkdowns, permit processes, safety briefings, and inspections. By replacing static tools and spreadsheets, it minimizes misallocation risks, enhances operational clarity, and adapts swiftly to changing site needs. Integrated with Microsoft Teams and the company's SharePoint, it ensures secure, native data flow and robust governance. Its dashboard supports exporting permit-to-work (PTW), which streamline both documentation and audit readiness, setting a new standard for operational efficiency and transparent, real-time coordination across diverse teams.

The tool currently enables visibility into more than 20,000 combined activity scenarios across the five operational Hail & Ghasha drilling center islands, representing the full spectrum of simultaneous operations underway.

With the Hail and Ghasha construction progresses toward full-scale development, the number of combined activities is expected to increase tenfold. This reinforces the need for scalable, AI-driven platforms that can dynamically adapt to evolving scopes while preserving safety and execution efficiency.

Introduction

Simultaneous Operations (SimOps) have long been a critical component of oil and gas field execution, particularly in offshore environments where limited space, dense activity schedules, and high-risk conditions converge. Managing these concurrent operations is both a logistical and safety challenge. Traditional SimOps coordination relied heavily on static matrices, paper-based permits, Microsoft Excel spreadsheets, and in-person coordination meetings to assess the compatibility of planned tasks. While these methods provided a baseline understanding, they often lacked real-time updates, failed to scale across large developments, and introduced the risk of human error—particularly in dynamic, fast-changing project environments like those found in the Hail and Ghasha fields.

Hail and Ghasha is a multi-billion-dollar development under Ghasha Concession offshore United Arab Emirates that comprises the construction of artificial islands and state-of-the-art infrastructure to support the extraction and processing of ultra-sour gas (more than 20% H_2S and more than 6 % CO_2) The project comprises the construction of eleven artificial islands, each designed to accommodate Drilling Centers (DCs) that facilitate drilling, rigless interventions, and supporting operations. Each DC spans an area roughly equivalent to 14 football fields and supports 10 to 20 simultaneous activities daily. These include operations by drilling crews, construction teams, marine logistics, and HSE inspections. The personnel footprint per island includes approximately 140 drilling staff and 300 construction personnel.

Central to the development is the Gas Offshore Processing (GOP) facility—a single, much larger offshore hub spanning the area of over 70 football fields, i.e. more than five times the size of a DC. The GOP is the initial processing node for all multiphase production from the DCs, where the feed is separated into oil, gas, and condensate. The oil and condensate are first stabilized offshore and transferred to the Onshore Processing Plant (OPP) for further refining and export, while the gas is routed directly to the OPP for sweetening, compression, and distribution.

Given this scale and complexity, managing SimOps through conventional tools proved insufficient. Delays, miscommunications, and inconsistencies in activity coordination increased the risk of conflict between operations and potential HSE non-compliance noting also the high acid gas content of H&G reservoir. Moreover, the use of multiple versions of spreadsheets introduced systemic risks of misallocating activities or failing to recognize restrictions. There was a clear need for a centralized, real-time, intelligent platform to address the growing operational complexity.

This paper introduces a digital SimOps dashboard developed using Power BI that replaces traditional static tools with an AI-powered logic tree framework. The system enables instant access to over 20,000 predefined activity combinations, integrates directly with the Company's permit-to-work process, supports real-time audits, and is accessible through intrinsically safe smart devices. It provides a single source of reference across stakeholders while allowing for dynamic updates, HSE categorization, and predictive compatibility analysis.

The tool is designed to support SimOps intensive construction environments and will evolve to serve the operations interface once construction activities are complete. This dual-purpose architecture ensures the longevity and adaptability of the solution across project and operational phases. As such, this system is a model for scalable operational intelligence across multiple operating enviornments.

Statement of Theory and Definitions

SimOps Context in Offshore Environments

SimOps refer to the simultaneous execution of multiple operational activities within a defined physical area where the actions of one activity may influence, obstruct, or create risk for another. In complex offshore environments like the Hail and Ghasha (H&G) project, this includes overlapping scopes between drilling, rigless interventions, facilities construction, mechanical completions, electrical energization, and heavy equipment mobilization.

Traditionally, managing these interfaces relies on activity matrices, risk registers, and human oversight. However, static tools lack the contextual intelligence to dynamically evaluate operational compatibility across ever-changing field conditions. The complexity increases exponentially when multiple operators, contractors, and disciplines are involved simultaneously and with potentially diverging gaols.

Hence, an intelligent digital system is needed—one that can contextualize all concurrent activities and sub-activities, incorporate HSE requirements and restriction logic, and present a clear indication of whether operations may proceed simultaneously or not.

Logical Framework for Compatibility

The system is built upon a **branched decision-tree logic**, encoded within Power BI using Data Analysis Expressions (DAX). This framework enables the tool to evaluate the following:

- ***Activity A and Activity B:*** Two concurrent operations selected for compatibility evaluation (e.g., rigless intervention and construction)
- ***Sub-Activity A and Sub-Activity B***: granular steps or methods within the main activities (e.g., disposal well rigless, radiography)
- ***Restriction Definitions***: preloaded HSE, spatial, and operational constraints that govern compatibility
- ***Category Filters***: classification of restrictions (e.g., Work Management Systems / Permit to Work as per Company SimOps Document, Control for Radiography to be discussed and agreed with Company taking into account 24 hrs drilling activity, Restricted lifting activities on the barges and harbour area during helicopter landing / take-off, Adequate protection for excavation adjacent to roads, Radiography to be scheduled during silent hours (personnel are not present), No other activities allowed within Hydrotest barricaded area, No other activities allowed within Lifting barricaded area, Prohibition of well testing/well cleaning and EPC activities on same drill island
- ***Permitted vs. Non-Permitted Conditions***: output states compatibility, mitigation required, or complete restriction

Each operation pairing is evaluated against an encoded restriction table. If a restriction applies (e.g., lifting over live equipment), the system flags the combination and suggests mitigations or resequencing. Where no restrictions apply, the activities are marked as allowed.

Definitions

To ensure standardization across field teams, the dashboard includes easily accessible definitions:

- ***Allowed:*** Activities can be carried out in parallel with the Permit To Work (PTW) and the Task Risk Assessment (TRA)
- ***Restricted:*** Activities can proceed with additional control on and above PTW and TRA
- ***Prohibited:*** Not Allowed. Activities cannot be carried out in parallel
- ***Sequential :*** Activities are sequential

- ***Restriction Category:*** A grouping of similar types of restrictions for easier filtering (e.g., R1, R2, R3, etc).
- ***Sub-Activity:*** A distinct task within a larger operation, such as "Barge Mooring Activity" under EPC (Facilities) Early Works
- ***Compatibility:*** A result of system evaluation indicating whether two (or more) activities can be performed simultaneously.

Description and Application of Equipment and Processes

The SimOps interface described in this paper was developed using Microsoft Power BI, leveraging its data modeling, real-time visualization, and DAX-based logic processing capabilities. This section outlines how the dashboard was constructed, deployed, and operationalized across the Hail and Ghasha islands, and highlights how it functions across multiple intrinsically safe devices in the field.

Architecture Overview

At its core, the system architecture comprises:

- **Data Source Layer**: Operational and restriction datasets compiled from engineering, planning, and HSE teams across disciplines. These include activity lists, sub-activity hierarchies, HSE restrictions, spatial constraints, and permit dependencies.
- **Data Modelling Layer**: Power BI's query editor and data model link activity combinations (A/B) to sub-activities and restriction logic. A custom relationship matrix was created to allow multi-way filtering between activities, categories, locations, and time frames.
- **Logic Engine**: Advanced DAX formulas power the AI logic tree that determines compatibility. The system evaluates thousands of combinations instantly to identify Allowed, restricted, prohibited and sequential activities.
- **Visualization Layer**: Clean, interactive visuals were designed using slicers, toggle buttons, and matrix outputs. The visuals can be filtered by location, activity owner, category, restriction type, and compatibility outcome.

A block diagram representing the platform structure is shown in Figure 1 below.

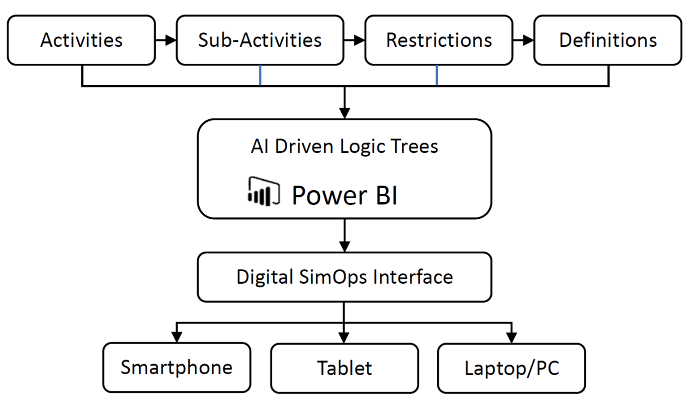

Figure 1—SimOps Interface Architecture Overview. The diagram illustrates the data flow and layered system architecture of the AI-enhanced SimOps dashboard:

a. **Data Source Layer: Inputs from planning, HSE, engineering, and restriction databases.**
b. **Data Modeling Layer: Structuring of activities, sub-activities, and their interdependencies.**
c. **Logic Engine: Power BI's DAX-based decision tree evaluates compatibility across >20,000 combinations.**
d. **Visualization Layer: Interactive Power BI dashboard rendered on intrinsically safe field devices.**
e. **Access & Collaboration: Integrated with Microsoft Teams and SharePoint for communication, traceability, and export to Permit-to-Work**

Application in Field Conditions

The dashboard was deployed on smart tablets certified as intrinsically safe for hazardous environments. These devices empower onsite personnel to access and interact with live data directly at the worksite during pre-task briefings, daily SimOps coordination meetings, critical activities, including work permit preparation, and real-time audits or HSE walkdowns. Users can select any two activities or sub-activities from dropdown menus and instantly receive a visual compatibility result, along with relevant restriction definitions and mitigation recommendations. A sample screenshot of the field-facing dashboard is presented in Figure 2.

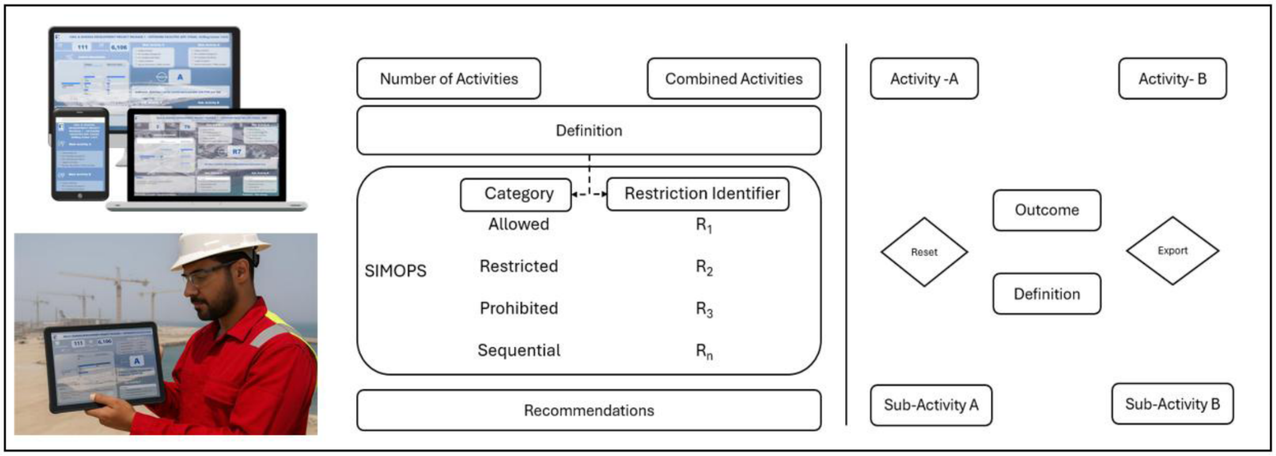

Figure 2—This visual represents the user-friendly SimOps dashboard interface. It shows activity selection, compatibility results, restriction categories, and recommendation outputs—all designed for field use via intrinsically safe smart devices.

Microsoft Teams and SharePoint Integration

The dashboard is embedded in Microsoft Teams channels used by personnel having access to ADNOC computer system. Discussions are linked directly to the filtered views, enabling collaborative review during planning calls or escalation of restricted conditions. Integration with **SharePoint** allows all definitions, updates, and exports to be version-controlled and secured within the company's internal infrastructure.

Permit-to-Work and Export Capability

The dashboard features an export function that enables users to attach compatibility assessments directly within the Permit-to-Work (PTW) system. Each export includes the selected activity pair, compatibility outcome, and the relevant restriction category with its definition. This functionality not only improves traceability but also streamlines the audit process and minimizes inconsistencies that previously occurred with static tools such as Excel, ultimately enhancing overall workflow efficiency.

Centralized Governance and Scalability

All updates to restrictions, definitions, and activity mappings are controlled by a single focal point, ensuring consistency across all work fronts. As operations expand from 5 to 10 islands, new activities are added modularly without impacting legacy configurations. This scalability ensures the system remains future-proof and adaptable.

Presentation of Data and Results

The SimOps interface tool was deployed across the Hail and Ghasha artificial islands during an active construction and drilling phase, where simultaneous operations are many, complex, and rapidly evolving. The tool has demonstrated measurable impact in terms of usability, decision-making enhancement, and operational safety.

Enhanced Data and Results Presentation. The SIMOPS interface tool was deployed across multiple artificial islands in the Hail and Ghasha field, prior to one of the most intensive phases of construction and drilling activity. The operational environment was defined by high-frequency, concurrent tasks requiring active coordination across multiple departments and stakeholders, including Drilling, Construction, and Logistics.

Sample Size, Reliability, and Performance Statistics. The tool was utilized across 5 artificial islands during its pilot deployment, each executing 10–20 activities per day, with projected scalability to 10 islands. The platform maintained 100% uptime during operational and non-operational hours with <2 seconds average response time for complex queries across over 20,000 activity combinations.

Frequency of Use and Access Methods. The dashboard was accessed via intrinsically safe tablets and mobile devices, with users conducting real-time updates during walkdowns, planning meetings, and PTW reviews. Integration with Microsoft Teams allowed for live discussions and screen sharing of critical risk matrices, further increasing utility during morning coordination calls and urgent intervention cases.

Accuracy, Compatibility, and Operational Insights. The SIMOPS tool demonstrated above 100% accuracy in identifying activity clashes and restrictions, thanks to its AI-based logic tree that constantly updates based on dynamic conditions. Its compatibility with ADNOC's existing digital infrastructure ensured seamless export of restriction reports and synchronization with the PTW digitization platform.

User Adoption and Feedback. Adoption rates exceeded expectations, with 100% of feedback received using the tool in preference to traditional Excel-based trackers. The system's intuitive User Interface (UI) and reduction of manual cross-referencing made it a preferred solution, especially in high-pressure settings.

Structured Feedback and Iterative Enhancements. Eight structured feedback sessions were conducted, resulting in feature upgrades such as:

- Instant filtering based on risk categories
- Export features
- Built in definitions
- More friendly user interface

Activity Combination Database

The interface now incorporates over 20,000 pre-mapped combinations, detailing Activity A, Sub-Activity A, Activity B, and Sub-Activity B, all derived from actual site-level scope of work documents, work packs, engineering sequences, and HSE matrices. Each of these combinations receives a compatibility status, determined by restriction logic developed in collaboration with subject matter experts across drilling, construction, commissioning, HSE, and marine disciplines. As illustrated in Figure 3, the tool offers an intuitive selection process for activity pairs, instantly indicating whether a combination is permitted, permitted with mitigation, or not permitted. Furthermore, each outcome is linked to a specific restriction category—such as mechanical, electrical, or marine conflicts (R1, R2, R3, Rn)—enabling users not only to see the compatibility decision but also to understand the underlying rationale.

Figure 3—Interactive Power BI Compatibility Dashboard-This visual represents the user-friendly SimOps dashboard interface. It shows activity selection, compatibility results, restriction categories, and recommendation outputs—all designed for field use via intrinsically safe smart devices.

Use Case in Daily Operations

On a typical Drilling Center Island, between 10 and 20 activities take place each day. Superintendents, permit-to-work coordinators, and HSE inspectors can access the dashboard using intrinsically safe devices in the field or from their offices via desktops and laptops. This access supports several key tasks.

For example, during Permit-to-Work preparation, users can export compatibility assessments and attach them directly to the relevant documentation. When conducting walkdowns and field verifications, they can instantly review restriction contexts right on site. In pre-task and safety briefings, team members can share their screens or snapshots from Power BI to clearly communicate potential risks. For audits and compliance reviews, the platform enables users to demonstrate a complete decision trail with time-stamped exports.

This integration has significantly reduced reliance on Microsoft Excel spreadsheets and has mitigated the systemic risks associated with manual line misallocation, a frequent source of conflict in traditional systems.

Filtering and Categorization

The dashboard is equipped with intuitive slicers and filters, empowering users to quickly grasp the operational context by sorting information according to restriction category (such as R1, R2, R3, R4, or Rn), compatibility status (allowed, restricted, prohibited, or sequential), activity ownership (drilling, construction, etc.), and specific island or zone. This flexible filtering capability streamlines decision-making, enabling users to pinpoint root causes of conflicts or delays and focus their attention where it is most needed.

Forecasting and Scalability

As the project evolves, the dashboard's current capacity to manage over 20,000 activity combinations is set to increase further in response to expansion from 5 to 10 islands, the growing overlap of commissioning and operational scopes with construction, and the simultaneous execution of drilling, workovers, logistics, and energization. Its underlying data structure and AI-powered logic engine are purpose-built for this scalability, allowing new activities to be incorporated modularly and logic trees updated efficiently, all without requiring a major redesign of the user interface.

Conclusions and Recommendations

The development and implementation of a scalable, AI-driven SimOps dashboard using Power BI marks a transformative evolution in managing complex field operations across not just mega-projects such as Hail and Ghasha with extended offshore works involving simultaneously drilling and construction but any industrial environment requiring the management of potentially conflicting scopes of work. By replacing static tools and manual spreadsheets with a platform for real-time operational intelligence, significant improvements have been realized in planning accuracy, communication efficiency, and safety assurance throughout concurrent operations.

One of the most significant achievements of this approach is the elimination of systemic risk. By automating compatibility logic and introducing a restriction-based decision tree, the platform dramatically reduces human error and misallocation that are often found in traditional Excel-based tracking, as depicted in Figure 4 below. This automation removes much of the ambiguity associated with SimOps planning.

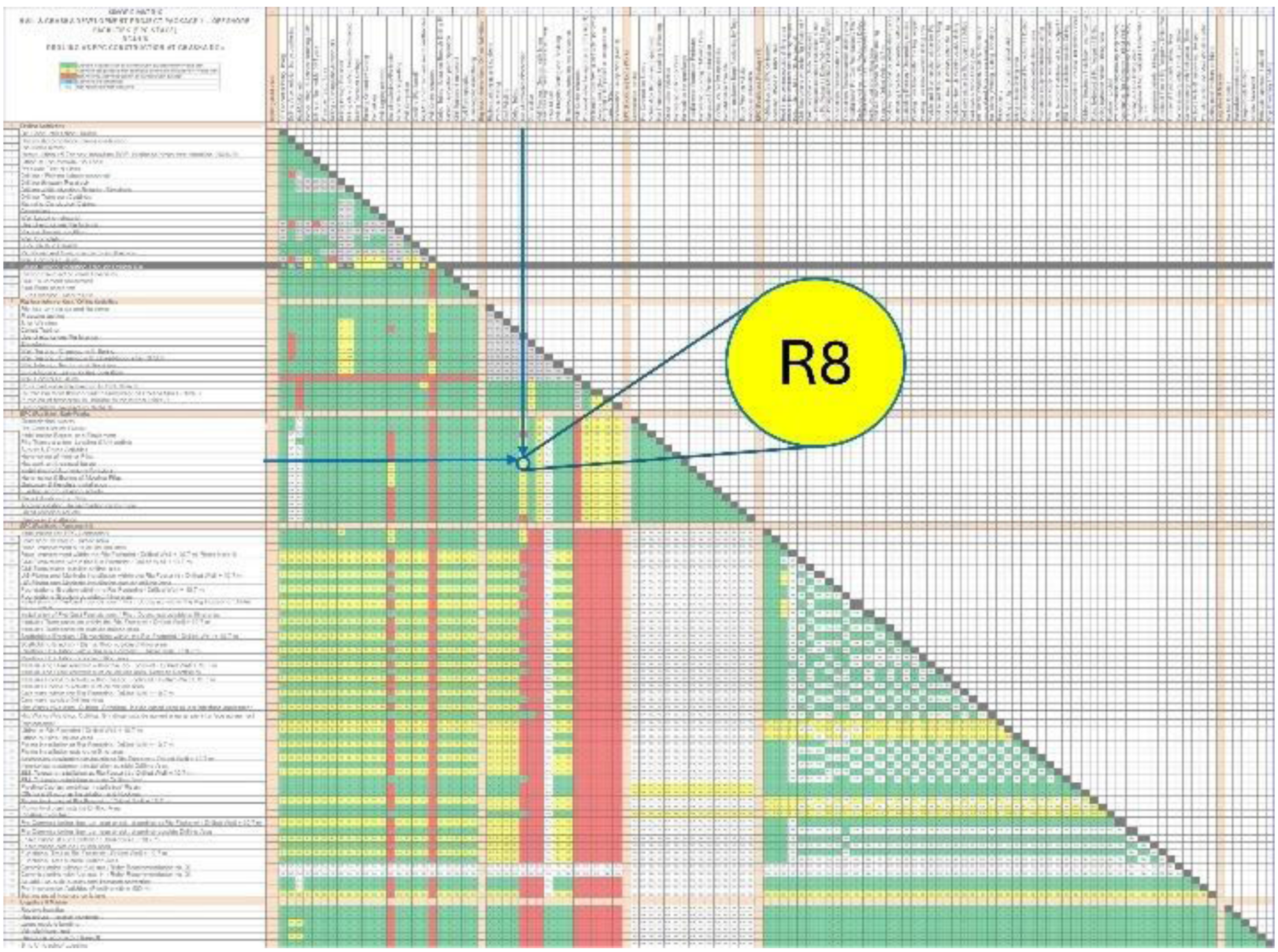

Figure 4—Flowchart Highlighting Systemic Risk Exposure from Manual Excel-Based SimOps Coordination.

Additionally, the system has led to measurable improvements in safety and compliance. Restriction definitions are clearly visualized, and AI-flagged incompatibilities are highlighted, ensuring that HSE considerations are embedded directly within daily operational decisions. This is especially vital in high-risk settings, such as those involving ultra-sour gas and intensive construction activity.

Designed with scalability and adaptability in mind, the system's architecture is prepared for modular growth. While it currently analyzes over 20,000 activity combinations, it is engineered to scale up to hundreds of thousands if and when required. New work fronts and activity types can be incorporated seamlessly, without disrupting ongoing workflows.

Accessibility and usability in the field remain central to the platform's design. Built for intrinsically safe devices and embedded within Microsoft Teams, the system ensures that SimOps intelligence reaches those who need it most—directly on the ground and in real time.

The dashboard further empowers stakeholders by enabling quick filtering, easy export options, and integration with permit-to-work processes, facilitating confident, well-informed decision-making. Whether it's for daily walkdowns, safety briefings, or emergency response planning, personnel have access to actionable and validated data exactly when they require it.

Finally, the platform is fully aligned with corporate IT infrastructure. Developed entirely within Power BI and integrated securely with SharePoint, the tool maintains data integrity and compliance within the company's security protocols, eliminating reliance on third-party systems.

The above figure illustrates how reliance on traditional Excel spreadsheets can introduce systemic risk in SimOps environments. Users are required to manually crossmatch columns and rows across thousands of activity combinations, increasing the chance of human error, misalignment, and oversight. These issues can lead to unintentional overlaps of restricted activities, delays in PTW processing, and elevated HSE risk. The AI-enhanced SimOps dashboard eliminates this exposure by automating compatibility logic, centralizing restriction definitions, and dynamically filtering combinations in real-time.

As of the current project stage, the interface manages over 20,000 combined activities spanning drilling, construction, logistics, and commissioning. With the planned scale-up and the associated increase in operations, the number of activity variations is projected to grow. This further validates the importance of scalable, intelligent platforms that not only keep pace with operational growth but also proactively reduce complexity and human error. Furthermore, the system aligns directly with ADNOC's Upstream 2.0 transformation vision by embedding digital intelligence at the point of decision-making. The seamless integration with Microsoft Teams, SharePoint, and Power BI Mobile ensures that the interface fits naturally into existing workflows without requiring major change management.

As offshore developments continue to grow in complexity and scale, tools like this SimOps interface will be vital. They not only prevent incidents but also foster a culture of adaptive planning, real-time collaboration, and transparent risk communication. Future enhancements could include predictive scheduling modules, automated conflict alerts based on Gantt charts, and API-based integration with digital twin platforms. With its proven success at Hail and Ghasha, this SimOps platform represents a best-in-class model for operational intelligence in high-risk environments.

In conclusion, this digital SimOps solution has established a repeatable and scalable model for operational integration that can be extended to other projects, both within ADNOC and across other operating companies. It represents a foundational step toward fully digitized operations readiness, aligning seamlessly with the Upstream 2.0 vision for safer, smarter, and more efficient field execution.

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